

Lab Experiment: 6

Title: Interfacing LCD with Arduino to Display Temperature using LM35 Sensor

Course title: IOT Laboratory

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Submitted to-

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1. Title

Interfacing LCD with Arduino to Display Temperature using LM35 Sensor

2. Objective

- To interface a **16x2 LCD** with Arduino
 - To read temperature data from **LM35 sensor**
 - To display real-time temperature on LCD
 - To understand analog input and display output
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3. Components Required

Component	Quantity
Arduino Uno	1
16x2 LCD Display	1
LM35 Temperature Sensor	1
Potentiometer (1k–10k)	1
Jumper wires	As required
Breadboard	1

4. Circuit Connection

LM35 Sensor Connection

- Pin 1 → **5V**
- Pin 2 → **A0 (Analog input)**
- Pin 3 → **GND**

 As shown in the diagram (page 1), LM35 provides analog voltage output

LCD Connection

- RS → Pin 12
- E → Pin 11
- D4–D7 → Pins 5, 4, 3, 2
- VSS → GND
- VDD → 5V
- V0 → Potentiometer (contrast control)

 The full wiring layout is illustrated in the circuit diagram (page 2)

5. Program (Arduino Code)

```
#include <LiquidCrystal.h>

int sensorpin = A0;
int temperature = 0;

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup() {
  pinMode(sensorpin, INPUT);
  lcd.begin(16, 2);
  lcd.print("Temperature:");
}

void loop() {
  temperature = analogRead(sensorpin);
  temperature = temperature * 0.48828125;

  lcd.setCursor(0,1);
  lcd.print(temperature);
  lcd.print((char)223); // degree symbol
  lcd.print("C");
}
```

6. Working Principle

- LM35 sensor outputs voltage proportional to temperature
- Arduino reads analog value (0–1023)
- Value is converted to temperature using formula:

$$\text{Temperature (}^{\circ}\text{C)} = \text{Analog Value} \times 0.48828125$$

- LCD displays temperature in real-time
 - Potentiometer controls LCD contrast
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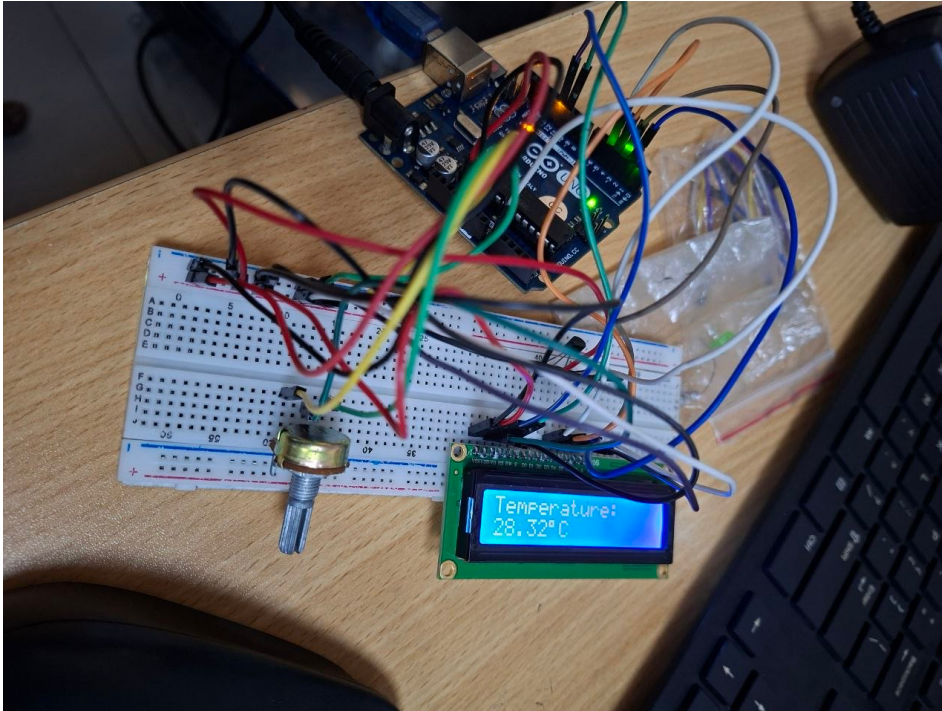
7. Procedure

1. Collect all required components
 2. Connect LM35 sensor and LCD as per diagram
 3. Connect Arduino to computer
 4. Open Arduino IDE
 5. Write or paste the code
 6. Select correct board and port
 7. Upload the program
 8. Observe temperature displayed on LCD
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8. Observation

- LCD displayed “Temperature:” on first line
- Second line showed real-time temperature value
- Value changed when temperature around sensor changed

9. Result



The temperature measured by LM35 sensor was successfully displayed on the LCD in real-time, demonstrating proper interfacing of sensor and display module with Arduino.

10. Applications

- Digital thermometers
 - Weather monitoring systems
 - Smart home automation
 - Industrial temperature monitoring
 - IoT display systems
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11. Precautions

- Ensure correct LCD pin connections
- Adjust potentiometer for proper display visibility
- Avoid loose wiring
- Use correct power supply
- Do not overheat LM35 sensor

12. Conclusion

This experiment demonstrated how to interface both an input device (LM35 sensor) and an output device (LCD) with Arduino. It provided a clear understanding of analog data acquisition and real-time display, which are essential concepts in embedded systems and IoT applications.